

# Dongseok Kwon

其他

## 基本信息

姓名: Dongseok Kwon

所属院校: 首尔国立大学

地区: South Korea

海外经历: 有

## 学科领域

微电子科学与工程

集成电路设计与集成系统

人工智能

电子与计算机工程

## 研究领域

微电子器件与制造

深度学习与神经网络

存储器设计与架构

低功耗电路与能量管理

薄膜材料与表面物理

低功耗设计与能耗优化

嵌入式系统与硬件设计

集成电路设计与制造

纳米电子与先进工艺

## 个人简介

Dongseok Kwon is affiliated with the Gwangju Institute of Science and Technology. His research focuses on Neuromorphic Devices, Semiconductor Devices, and Gas Sensors. He has published extensively in these areas, contributing to advancements in neuromorphic computing, hardware-based deep neural networks, and the integration of NAND flash memory cells as synaptic devices. His work also includes studies on adaptive learning rules, weight quantization methods, and the development of reconfigurable field-effect transistors for neural networks. Kwon's research is widely recognized in leading journals and conferences.

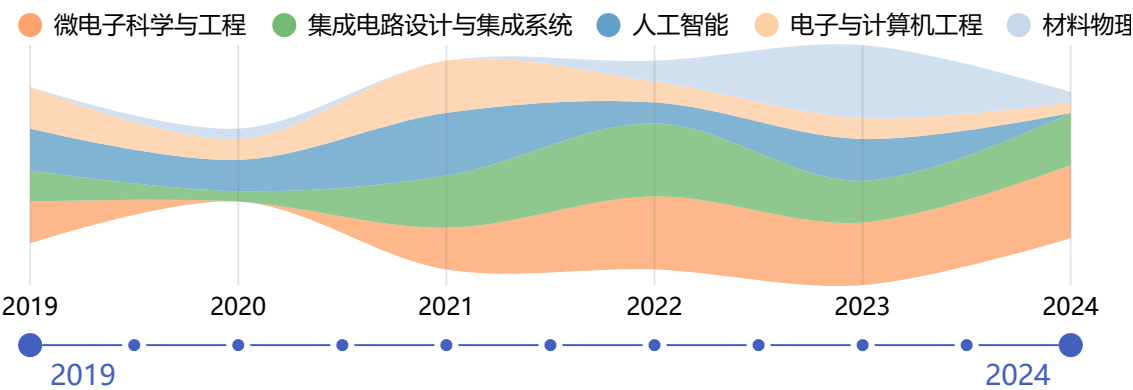
## 工作经历

教育经历

Seoul National University - Master

2017-03-02 -

近年发表的论文



Robust 1/f Noise Unaffected by Program/Erase Cycling-Induced Damage in Ferroelectric Schottky Barrier FETs  
IEEE ELECTRON DEVICE LETTERS  
Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho  
被引用次数: 1 2024

NOR-Type Flash Array Based on Four-Terminal TFT Synaptic Devices Capable of Selective Program/Erase Exploiting Fowler-Nordheim Tunneling  
IEEE ELECTRON DEVICE LETTERS  
Hwang, Joon; Park, Min-Kyu; Kang, Won-Mook; Koo, Ryun-Han; Lee, Kyu-Ho; Kwon, Dongseok; Choi, Woo Young; Lee, Jong-Ho  
被引用次数: 0 2024

Toward Optimized In-Memory Reinforcement Learning: Leveraging 1/f Noise of Synaptic Ferroelectric Field-Effect-Transistors for Efficient Exploration  
ADVANCED INTELLIGENT SYSTEMS  
Kim, Jangsaeng; Shin, Wonjun; Yim, Jiyong; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho  
被引用次数: 2 2024

SNNSim: Investigation and Optimization of Large-Scale Analog Spiking Neural Networks Based on Flash Memory Devices  
ADVANCED INTELLIGENT SYSTEMS  
Ko, Jong Hyun; Kwon, Dongseok; Hwang, Joon; Lee, Kyu-Ho; Oh, Seongbin; Kim, Jeonghyun; Im, Jiseong; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong-Ho  
被引用次数: 1 2024

## Stochasticity in ferroelectric memory devices with different bottom electrode crystallinity

CHAOS SOLITONS & FRACTALS

Koo, Ryun-Han; Shin, Wonjun; Jung, Gyuweon; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 4

2024

## On-Chip Learning in Vertical NAND Flash Memory Using Forward-Forward Algorithm

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Sung-Ho; Ko, Jonghyun; Lee, In-Seok; Koo, Ryun-Han; Kim, Jeong-Hyun; Yang, Yeongheon; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 1

2024

## Demonstration of In-Memory Biosignal Analysis: Novel High-Density and Low-Power 3D Flash Memory Array for Arrhythmia Detection

ADVANCED SCIENCE

Kim, Jangsaeng; Im, Jiseong; Shin, Wonjun; Lee, Soochang; Oh, Seongbin; Kwon, Dongseok; Jung, Gyuweon; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 1

2024

## Proposition of Adaptive Read Bias: A Solution to Overcome Power and Scaling Limitations in Ferroelectric-Based Neuromorphic System

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Seungwhan; Im, Jiseong; Park, Sung-Ho; Ko, Jong Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2024

## Si-Based Dual-Gate Field-Effect Transistor Array for Low-Power On-Chip Trainable Hardware Neural Networks

ADVANCED INTELLIGENT SYSTEMS

Lee, Kyu-Ho; Kwon, Dongseok; Lee, In-Seok; Hwang, Joon; Im, Jiseong; Bae, Jong-Ho; Choi, Woo Young; Woo, Sung Yun; Lee, Jong-Ho

被引用次数: 3

2024

## Effect of Carrier Transport Process on Tunneling Electroresistance in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Dae Hwan; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 16

2023

## Low-frequency noise characteristics of indium-gallium-zinc oxide ferroelectric thin-film transistors with metal-ferroelectric-metal-insulator-semiconductor structure

APPLIED PHYSICS LETTERS

Shin, Wonjun; Park, Eun Chan; Koo, Ryun-Han; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 3

2023

## Analog Synaptic Devices Based on IGZO Thin-Film Transistors with a Metal-Ferroelectric-Metal-Insulator-Semiconductor Structure for High-Performance Neuromorphic Systems

ADVANCED INTELLIGENT SYSTEMS

Kwon, Dongseok; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Hwang, Joon; Bae, Jong-Ho; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2023

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## 1/f Noise in Synaptic Ferroelectric Tunnel Junction: Impact on Convolutional Neural Network

ADVANCED INTELLIGENT SYSTEMS

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 11

2023

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## Demonstration of Pulse Width Modulation Function Using Single Positive Feedback Device for Neuron

IEEE ELECTRON DEVICE LETTERS

Woo, Sung Yun; Kwon, Dongseok; Park, Byung-Gook; Lee, Jong-Ho; Bae, Jong-Ho

被引用次数: 2

2023

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## Unveiling Resistance Switching Mechanisms in Undoped HfO<sub>x</sub> Ferroelectric Tunnel Junction Using Low-Frequency Noise Spectroscopy

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 9

2023

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## Channel-Length-Dependent Low-Frequency Noise Characteristics of Ferroelectric Junctionless Poly-Si Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kim, Sangwoo; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Deok-Hwang; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2023

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## Optimizing Post-Metal Annealing Temperature Considering Different Resistive Switching Mechanisms in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 10

2023

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## Unraveling Threshold Voltage Instability in Ferroelectric Junctionless FETs Using Low-Frequency Noise Measurement With Base Bias

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

## Comparative Analysis of n- and p-Type Ferroelectric Tunnel Junctions Through Understanding of Non-FE Resistance Switching

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Lee, Kyungmin; Park, Sung-Ho; Im, Jiseong; Ko, Jong-Hyun; Kim, Jeong-Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 6

2023

## Fast-response/recovery In<sub>2</sub>O<sub>3</sub> thin-film transistor-type NO<sub>2</sub> gas sensor with floating-gate at low temperature

SENSORS AND ACTUATORS B-CHEMICAL

Yeom, Gyuho; Kwon, Dongseok; Shin, Wonjun; Park, Min-Kyu; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 10

2023

## Efficient Hybrid Training Method for Neuromorphic Hardware Using Analog Nonvolatile Memory

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Kwon, Dongseok; Woo, Sung Yun; Hwang, Joon; Kim, Hyeongsu; Bae, Jong-Ho; Shin, Wonjun; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 0

2023

## Hardware-Based Ternary Neural Network Using AND-Type Poly-Si TFT Array and Its Optimization Guideline

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kwon, Dongseok; Park, Min-Kyu; Kang, Won-Mook; Hwang, Joon; Koo, Ryun-Han; Bae, Jong-Ho; Lee, Jong-Ho

被引用次数: 2

2023

## Self-Curable Synaptic Ferroelectric FET Arrays for Neuromorphic Convolutional Neural Network

ADVANCED SCIENCE

Shin, Wonjun; Im, Jiyong; Koo, Ryun-Han; Kim, Jaehyeon; Kwon, Ki-Ryun; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho; Kwon, Daewoong

被引用次数: 21

2023

## Energy Efficient Artificial Olfactory System with Integrated Sensing and Computing Capabilities for Food Spoilage Detection

ADVANCED SCIENCE

Jung, Gyuweon; Kim, Jaehyeon; Hong, Seongbin; Shin, Hunhee; Jeong, Yujeong; Shin, Wonjun; Kwon, Dongseok; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 8

2023

## Reconfigurable neuromorphic computing block through integration of flash synapse arrays and super-steep neurons

SCIENCE ADVANCES

Kwon, Dongseok; Woo, Sung Yun; Lee, Kyu-Ho; Hwang, Joon; Kim, Hyeongsu; Park, Sung-Ho; Shin, Wonjun; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 8

2023

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## Voltage scheme for string-select transistors to improve inhibition characteristics during 1-bit erase in vertical NAND flash

APPLIED PHYSICS LETTERS

Park, Sung-Ho; Yoo, Ho-Nam; Yang, Yeongheon; Back, Jong-Won; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 2

2023

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## Super-steep synapses based on positive feedback devices for reliable binary neural networks

APPLIED PHYSICS LETTERS

Kwon, Dongseok; Kim, Hyeongsu; Lee, Kyu-Ho; Hwang, Joon; Shin, Wonjun; Bae, Jong-Ho; Woo, Sung Yun; Lee, Jong-Ho

被引用次数: 2

2023

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## Effects of temperature and DC cycling stress on resistive switching mechanisms in hafnia-based ferroelectric tunnel junction

APPLIED PHYSICS LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwak, Been; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2023

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## Highly Efficient Self-Curing Method in MOSFET Using Parasitic Bipolar Junction Transistor

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Kwon, Dongseok; Hwang, Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 8

2022

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## Investigation of Low-Frequency Noise Characteristics of Ferroelectric Tunnel Junction: From Conduction Mechanism and Scaling Perspectives

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kwon, Dongseok; Koo, Ryun-Han; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 17

2022

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## Effects of High-Pressure Annealing on the Low-Frequency Noise Characteristics in Ferroelectric FET

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Sihyun; Lee, Kitae; Kwon, Dongseok; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 40

2022

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## Highly Linear Analog Spike Processing Block Integrated With an AND-Type Flash Array and CMOS Neuron Circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kyu-Ho; Kwon, Dongseok; Woo, Sung Yun; Ko, Jong Hyun; Choi, Woo Young; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 7

2022

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## CMOS-Compatible Low-Power Gated Diode Synaptic Device for Hardware-Based Neural Network

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Min-Kyu; Yoo, Ho-Nam; Hwang, Joon; Woo, Sung Yun; Kwon, Dongseok; Seo, Young-Tak; Lee, Jong-Ho; Bae, Jong-Ho

被引用次数: 5

2022

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## Review of Analog Neuron Devices for Hardware-based Spiking Neural Networks

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kwon, Dongseok; Woo, Sung Yun; Lee, Jong-Ho

被引用次数: 8

2022

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## Retention Improvement in Vertical NAND Flash Memory Using 1-bit Soft Erase Scheme and its Effects on Neural Networks

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Park, Sung-Ho; Kwon, Dongseok; Yoo, Ho-Nam; Back, Jong-Won; Hwang, Joon; Yang, Yeongheon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 11

2022

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## Demonstration of integrate-and-fire neuron circuit for spiking neural networks

SOLID-STATE ELECTRONICS

Woo, Sung Yun; Kang, Won-Mook; Seo, Young-Tak; Lee, Soochang; Kwon, Dongseok; Oh, Seongbin; Bae, Jong-Ho; Lee, Jong-Ho

被引用次数: 4

2022

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## Synergistic improvement of sensing performance in ferroelectric transistor gas sensors using remnant polarization

MATERIALS HORIZONS

Shin, Wonjun; Yim, Jiyong; Bae, Jong-Ho; Lee, Jung-Kyu; Hong, Seongbin; Kim, Jaehyeon; Jeong, Yujeong; Kwon, Dongseok; Koo, Ryun-Han; Jung, Gyuweon; Han, Changhyeon; Kim, Jeonghan; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 19

2022

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## Comprehensive and accurate analysis of the working principle in ferroelectric tunnel junctions using low-frequency noise spectroscopy

NANOSCALE

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Yim, Jiyong; Kwon, Dongseok; Kim, Yeonwoo; Yu, Junsu; Hwang, Joon; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 29

2022

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## Neuron Circuits for Low-Power Spiking Neural Networks Using Time-To-First-Spike Encoding

IEEE ACCESS

Oh, Seongbin; Kwon, Dongseok; Yeom, Gyuho; Kang, Won-Mook; Lee, Soochang; Woo, Sung Yun; Kim, Jaehyeon; Lee, Jong-Ho

被引用次数: 13

2022

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### On-Chip Trainable Spiking Neural Networks Using Time-To-First-Spike Encoding

IEEE ACCESS

Im, Jiseong; Kim, Jaehyeon; Yoo, Ho-Nam; Baek, Jong-Won; Kwon, Dongseok; Oh, Seongbin; Kim, Jangsaeng; Hwang, Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 2

2022

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### Novel, parallel and differential synaptic architecture based on NAND flash memory for high-density and highly-reliable binary neural networks

NEUROCOMPUTING

Lee, Sung-Tae; Kim, Hyeongsu; Yoo, Honam; Kwon, Dongseok; Lee, Jong-Ho

被引用次数: 2

2022

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### Efficient fusion of spiking neural networks and FET-type gas sensors for a fast and reliable artificial olfactory system

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Kim, Jaehyeon; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 26

2021

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### 3-D AND-Type Flash Memory Architecture With High- $\kappa$ Gate Dielectric for High-Density Synaptic Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Seo, Young-Tak; Kwon, Dongseok; Noh, Yoohyun; Lee, Soochang; Park, Min-Kyu; Woo, Sung Yun; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 24

2021

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### Hardware-Based Spiking Neural Networks Using Capacitor-Less Positive Feedback Neuron Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kwon, Dongseok; Woo, Sung Yun; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 2

2021

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### Investigation of Low-Frequency Noise Characteristics in Gated Schottky Diodes

IEEE ELECTRON DEVICE LETTERS

Kwon, Dongseok; Shin, Wonjun; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 12

2021

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### Impacts of Program/Erase Cycling on the Low-Frequency Noise Characteristics of Reconfigurable Gated Schottky Diodes

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kwon, Dongseok; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho

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## Hardware -based spiking neural network architecture using simplified backpropagation algorithm and homeostasis functionality

NEUROCOMPUTING

Kim, Jangsaeng; Kwon, Dongseok; Woo, Sung Yun; Kang, Won-Mook; Lee, Soochang; Oh, Seongbin; Kim, Chul-Heung; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 19

2021

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## On-chip trainable hardware-based deep Q-networks approximating a backpropagation algorithm

NEURAL COMPUTING & APPLICATIONS

Kim, Jangsaeng; Kwon, Dongseok; Woo, Sung Yun; Kang, Won-Mook; Lee, Soochang; Oh, Seongbin; Kim, Chul-Heung; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 8

2021

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## Low-power and reliable gas sensing system based on recurrent neural networks

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

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2021

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## Spiking Neural Networks With Time-to-First-Spike Coding Using TFT-Type Synaptic Device Model

IEEE ACCESS

Oh, Seongbin; Lee, Soochang; Woo, Sung Yun; Kwon, Dongseok; Im, Jiseong; Hwang, Joon; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 5

2021

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## Hardware-Based Spiking Neural Network Using a TFT-Type AND Flash Memory Array Architecture Based on Direct Feedback Alignment

IEEE ACCESS

Kang, Won-Mook; Kwon, Dongseok; Woo, Sung Yun; Lee, Soochang; Yoo, Honam; Kim, Jangsaeng; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 12

2021

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## Direct Gradient Calculation: Simple and Variation-Tolerant On-Chip Training Method for Neural Networks

ADVANCED INTELLIGENT SYSTEMS

Kim, Hyungyo; Hwang, Joon; Kwon, Dongseok; Kim, Jangsaeng; Park, Min-Kyu; Im, Jiseong; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 6

2021

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## Proposition of deposition and bias conditions for optimal signal-to-noise-ratio in resistor- and FET-type gas sensors

NANOSCALE

Shin, Wonjun; Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Jang, Dongkyu; Kwon, Dongseok; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 34

2020

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## Low-Power and High-Density Neuron Device for Simultaneous Processing of Excitatory and Inhibitory Signals in Neuromorphic Systems

IEEE ACCESS

Woo, Sung Yun; Kwon, Dongseok; Choi, Nagyong; Kang, Won-Mook; Seo, Young-Tak; Park, Min-Kyu; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 14

2020

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## NAND Flash Based Novel Synaptic Architecture for Highly Robust and High-Density Quantized Neural Networks With Binary Neuron Activation of (1,0)

IEEE ACCESS

Lee, Sung-Tae; Kwon, Dongseok; Kim, Hyeongsu; Yoo, Honam; Lee, Jong-Ho

被引用次数: 15

2020

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## On-Chip Training Spiking Neural Networks Using Approximated Backpropagation With Analog Synaptic Devices

FRONTIERS IN NEUROSCIENCE

Kwon, Dongseok; Lim, Suhwan; Bae, Jong-Ho; Lee, Sung-Tae; Kim, Hyeongsu; Seo, Young-Tak; Oh, Seongbin; Kim, Jangsaeng; Yeom, Kyuho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 43

2020

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## Efficient precise weight tuning protocol considering variation of the synaptic devices and target accuracy

NEUROCOMPUTING

Kim, Hyeongsu; Bae, Jong-Ho; Lim, Suhwan; Lee, Sung-Tae; Seo, Young-Tak; Kwon, Dongseok; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 15

2020

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## Emerging memory technologies for neuromorphic computing

NANOTECHNOLOGY

Kim, Chul-Heung; Lim, Suhwan; Woo, Sung Yun; Kang, Won-Mook; Seo, Young-Tak; Lee, Sung-Tae; Lee, Soochang; Kwon, Dongseok; Oh, Seongbin; Noh, Yoohyun; Kim, Hyeongsu; Kim, Jangsaeng; Bae, Jong-Ho; Lee, Jong-Ho

被引用次数: 72

2019

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## Reconfigurable Field-Effect Transistor as a Synaptic Device for XNOR Binary Neural Network

IEEE ELECTRON DEVICE LETTERS

Bae, Jong-Ho; Kim, Hyeongsu; Kwon, Dongseok; Lim, Suhwan; Lee, Sung-Tae; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 30

2019

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## Adaptive learning rule for hardware-based deep neural networks using electronic synapse devices

## NEURAL COMPUTING & APPLICATIONS

Lim, Suhwan; Bae, Jong-Ho; Eum, Jai-Ho; Lee, Sungtae; Kim, Chul-Heung; Kwon, Dongseok; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 58

2019

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## Adaptive Weight Quantization Method for Nonlinear Synaptic Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kwon, Dongseok; Lim, Suhwan; Bae, Jong-Ho; Lee, Sung-Tae; Kim, Hyeongsu; Kim, Chul-Heung; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 33

2019

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## Operation Scheme of Multi-Layer Neural Networks Using NAND Flash Memory as High-Density Synaptic Devices

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Lee, Sung-Tae; Lim, Suhwan; Choi, Nag Yong; Bae, Jong-Ho; Kwon, Dongseok; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 32

2019

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## Highly Reliable Inference System of Neural Networks Using Gated Schottky Diodes

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Lim, Suhwan; Kwon, Dongseok; Eum, Jai-Ho; Lee, Sung-Tae; Bae, Jong-Ho; Kim, Hyeongsu; Kim, Chul-Heung; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 15

2019

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## Near-Linear Potentiation Mechanism of Gated Schottky Diode as a Synaptic Device

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Bae, Jong-Ho; Lim, Suhwan; Kwon, Dongseok; Eum, Jai-Ho; Lee, Sung-Tae; Kim, Hyeongsu; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 4

2019

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## Synaptic device using a floating fin-body MOSFET with memory functionality for neural network

SOLID-STATE ELECTRONICS

Woo, Sung Yun; Choi, Kyu-Bong; Lim, Suhwan; Lee, Sung-Tae; Kim, Chul-Heung; Kang, Won-Mook; Kwon, Dongseok; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

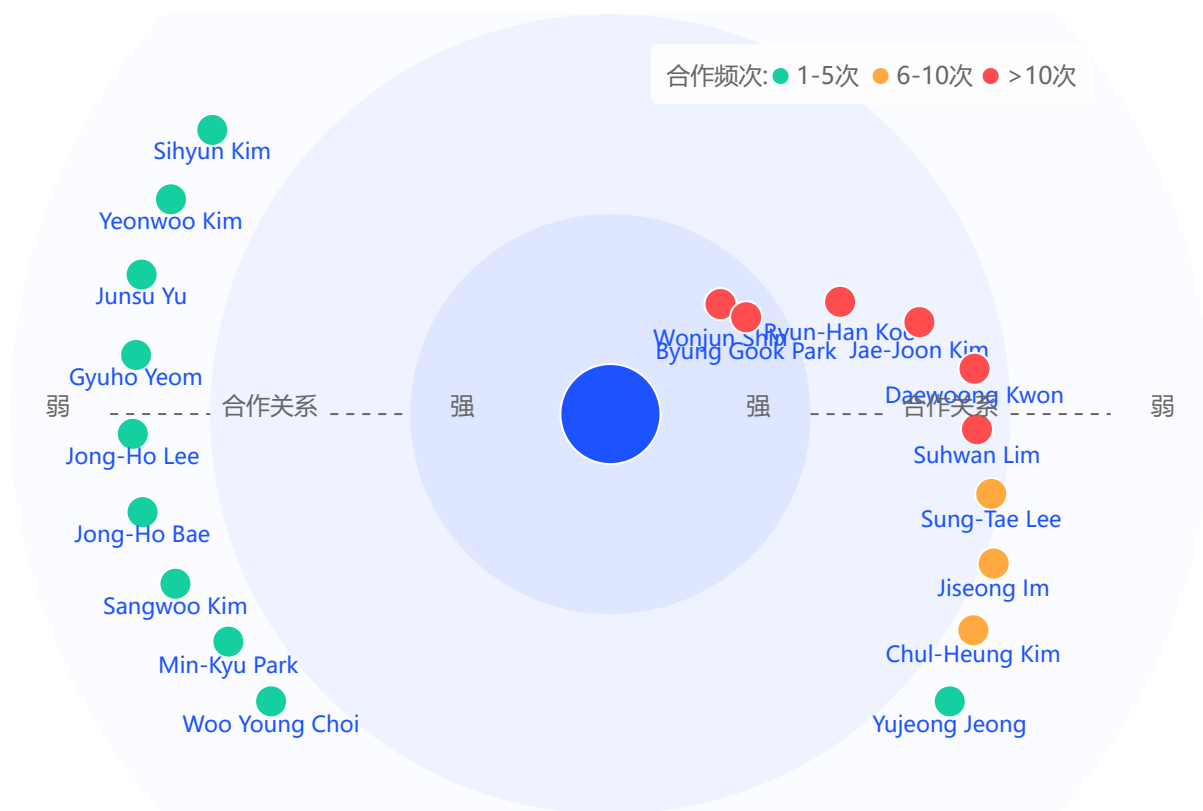
被引用次数: 5

2019

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## 学术合作

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Wonjun Shin

成均馆大学

合作次数: 31

Byung Gook Park

首尔国立大学

合作次数: 30

Ryun-Han Koo

首尔国立大学

合作次数: 22

Jae-Joon Kim

首尔国立大学

合作次数: 16

Daewoong Kwon

汉阳大学

合作次数: 12

还有 14 个合作者